02 · Case L: Jack's noise-induced hearing loss

The words of this track, as it says them. Bold brown is the examiner.

Case L. Jack's noise-induced hearing loss. Six marks.

Jack works in a very noisy open-plan office, and for years he has listened to loud music to drown it out. Now he has trouble hearing and understanding. Which parts of the ear are damaged, and where are they? How does loud noise over time damage them, and how does that lead to deafness? What kind of deafness is this, and how does it compare with the other kind?

This is one of the two case studies she names for Test 3, so treat it as close to guaranteed. It's four questions in one. Answer them in order, and that's your six marks.

The hook: Corti, snap, silent, sensorineural. The hairs in the organ of Corti snap off, the nerve goes silent, and that's sensorineural deafness.

First: which parts are damaged, and where? The hair cells. They're the mechanoreceptors for hearing, and they sit in the organ of Corti. Be specific, because the case asks you to be. The organ of Corti sits on the basilar membrane, in the cochlear duct, which is the scala media, inside the cochlea, in the inner ear. Hair cells, organ of Corti, basilar membrane, cochlea, inner ear. Give the whole address.

Second: how does loud noise damage them? Loud sound is a wave with a big amplitude. The basilar membrane moves so hard that the hairs bend against the tectorial membrane, over and over, until they break off. Her deck gives you two numbers. Continuous exposure to eighty-five decibels or more does it over time. A hundred and thirty decibels or more does it at once. Jack is the over-time kind: years of noise.

Third: how does that cause deafness? A hair cell's job is transduction: turning vibration into nerve impulses. A broken hair cell can't do it. So no impulses travel along the vestibulocochlear nerve, cranial nerve eight, to the auditory cortex in the temporal lobe. And hair cells don't grow back. So noise-induced hearing loss is permanent.

Fourth: what kind of deafness, and the other kind? This is sensorineural deafness: damage anywhere from the hair cells of the cochlea to the auditory cortex. It's usually irreversible. Depending on where the damage is, a cochlear implant can help. The other kind is conduction deafness. Sound is blocked before it reaches the inner ear, somewhere in the outer or middle ear. Impacted earwax. A perforated eardrum. A middle ear infection, like glue ear. Conduction deafness is often reversible.

Where the marks go missing. One: the semicircular canals have hair cells too, but those are for balance. Her own question offers them as the wrong answer. Jack's broken hair cells are in the cochlea. Two: the sound came in through a perfectly normal ear canal, so people write conduction. It isn't. Nothing blocked the sound. The damage is in the inner ear, so it's sensorineural. Three: don't stop at "the cochlea". Name the organ of Corti, and where it sits.

Once more. Jack's hearing loss. What's damaged, how, why he's deaf, and what kind.

The hair cells of the organ of Corti, on the basilar membrane, in the cochlea of the inner ear. Loud, high-amplitude sound bends the hairs against the tectorial membrane until they break off: eighty-five decibels over time, a hundred and thirty at once. No impulses go down nerve eight to the brain, and hair cells don't grow back, so it's permanent. That's sensorineural deafness. The other kind, conduction, is a blockage in the outer or middle ear, like earwax or glue ear, and it's often reversible.